

Video Analysis: Identifying Inaccessible Brands and Accessibility Questions from Conversations Between Visually Impaired Individuals and their Sighted Interpreters

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Abstract

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Video Analysis: Identifying Inaccessible Brands and Accessibility Questions from Conversations Between Visually Impaired Individuals and their Sighted Interpreters.

Thesis directed by Assistant Professor Dr. Danna Gurari.

This goal is to understand the challenges faced by visually impaired individuals in their daily lives. To do so, we analyze video conversations between a sighted volunteer and a visually impaired user that include both visual and audio modalities. Two pipelines are developed: a video pipeline that extracts information on brands of products that are difficult for visually impaired individuals to access, and an audio pipeline that captures the conversation between the volunteer and the user to gain insights into the types of inaccessibility issues that visually impaired users face. Various computer vision, audio, and text processing methods are employed for this analysis. Results are analyzed quantitatively using metrics such as precision, recall, and F1 score. We discovered that a combination of the video and audio analysis pipelines outperforms relying on the video pipeline alone. A qualitative analysis of the audio pipeline also revealed that it extracts all questions asked by the user and the corresponding responses of the volunteer in most cases. In closing, this work presents an efficient way to analyze videos of conversations between visually impaired individuals and volunteers, enabling the identification of the specific brands and types of issues that visually impaired individuals face in their daily lives.

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Chapter 1

Introduction

1.1 Motivation

The World Health Organization states that more than one billion individuals worldwide are affected by vision impairment, which can include varying degrees of distance vision impairment or complete blindness (54). Unfortunately, many everyday products are not designed to address the needs of individuals whose eye sight cannot be improved through surgery or corrective lenses. This may be in part due to manufacturers lacking awareness of the challenges that these visually impaired individuals face when using such products.

Accessibility challenges individuals with visual impairments encounter include reading cooking instructions, reading labels on food cans, identifying colors of objects, and setting the temperature for air conditioners. These may seem like simple tasks to someone with total vision, but for those with visual impairments, these tasks can be daunting and may require assistance from a sighted individual.

To address such issues, various tools and systems have been developed to assist visually impaired individuals. These range from connecting visually impaired individuals with sighted volunteers who provide assistance, to the use of automated systems that provide assistance through technology. Although visual assistance tools

are available, they have certain limitations and may not be suitable for all situations. For instance, when it comes to low-light conditions, most visual assistance tools face difficulty in analyzing images or videos. Likewise, in intricate settings such as a congested street or a messy table, it can be difficult to comprehend what is happening. Moreover, in some cases, people may require assistance with personal matters like operating an ATM, but they may not feel comfortable sharing private information.

The long-term objective of this research is to design a scalable solution for identifying accessibility issues for individuals with visual impairment. To achieve this, we wanted to design an automated framework to analyze video recordings of conversations between visually impaired individuals and sighted volunteers who are answering visual questions. In analyzing these conversations, we aimed to identify the specific products and brands that visually impaired individuals are sharing in their video feeds and so are likely inaccessible. We also aimed to analyze the questions asked by visually impaired users in order to gain insights into the specific accessibility issues that users are experiencing.

Altogether, we aim to reduce the accessibility barriers that individuals with visual impairments face in their daily lives. This, in turn, would promote greater independence and social inclusion for this population. The aforementioned information from video conversations can inform what is needed to make our environments more inclusive and accessible.

1.2 Problem Scope

Towards automatically analyzing videos of conversations between visually impaired individuals and sighted volunteers, our approach needs to be multi-modal, meaning there should be analysis of both the audio and video. Consequently, two separate pipelines are needed to analyze each modality. One pipeline is responsible for analyzing the visual aspects of the videos in order to identify inaccessible content, such as the brands and products. The second pipeline is responsible for analyzing the

audio aspects of the videos to understand what types of questions and answers are being asked by the individuals engaged in the conversations. Finally, a synthesis step is needed to analyze both pipelines in conjunction. Our research aims to address four key challenges in this field: determining relevant videos for analysis, reducing the cost of video analysis, comprehending unstructured conversations, and fusing information from video and audio pipelines

Our first challenge centers on determining which videos are relevant for analysis to identify inaccessible brands. This centers on determining if the object in question is a product and that the particular brand of the product is mentioned by either the visually impaired caller or the sighted volunteer who serves as a visual interpreter. Within our dataset of 97 videos, as will be described, we found only 15 of them align with this problem scope.

Our second challenge stems from the fact that videos included in our dataset are typically recorded at 25-30 frames per second and have an average length of 3-5 minutes, resulting in a vast number of frames to be analyzed. One of the significant challenges associated with analyzing these videos is the high cost, as analyzing every frame in a video is time-consuming and computationally expensive. Observing that many video frames are not relevant, such as segments with a lot of camera movement when the user attempts to adjust the camera’s positioning to provide the volunteer with a better view of their surroundings, this presents an opportunity for us to develop more efficient methods for analyzing videos. By detecting regions of interest, or key-frames, in a video, we can reduce the number of frames to analyze without sacrificing the accuracy of our analysis.

The third challenge we face is comprehending the inquiries posed by the user and the corresponding answers provided by the volunteer, with the aim of obtaining an understanding of the kind of problems encountered by individuals with visual impairments. We aim to do this for all the videos in our dataset, irrespective of the task at hand. The challenge that arises is that the conversations are often unstructured, making it difficult to discern who is asking the questions and what the

corresponding responses are. This presents an opportunity for us to develop a pipeline that can identify each speaker in the conversation. By modeling the conversation into dialogues, we can analyze the inherent patterns in these dialogues and extract relevant information from the conversation. Additionally, this approach can give us a better understanding of specific segments in the video where the questions about accessibility issues are being asked, allowing us to analyze these segments more efficiently.

Our fourth challenge involves the fusion of information from the video and audio pipelines. By utilizing both the video and audio content, we can focus our analysis on specific segments within the video that hold particular relevance to the task we aim to accomplish. This targeted approach is valuable, as not all portions of a video may contain information that is pertinent or directly related to the task at hand. Additionally, leveraging the distinct information of both video and audio modalities can lead to a more comprehensive understanding of the content being examined.

1.3 Idea

The central concept of this thesis revolves around extracting distinct information from two separate modalities - video and audio. Through the video pipeline, our objective is to identify and extract brands that are not accessible, while the audio pipeline focuses on extracting questions and their corresponding answers by analyzing conversational patterns.

Our proposed methodology entails the development and utilization of two primary, independent pipelines designed to efficiently analyze and extract pertinent information from the given video content. These pipelines are summarized in Figure 1.1. The first pipeline, which emphasizes the visual elements of the video without considering the audio, employs a range of computer vision techniques to effectively recognize and extract any brand present within the video content. The second pipeline concentrates on examining the audio aspect of the video, with the aim of extracting any questions posed during the video, as well as their associated

answers. This is achieved by identifying conversational structures and capitalizing on the patterns that typically emerge in dialogues.

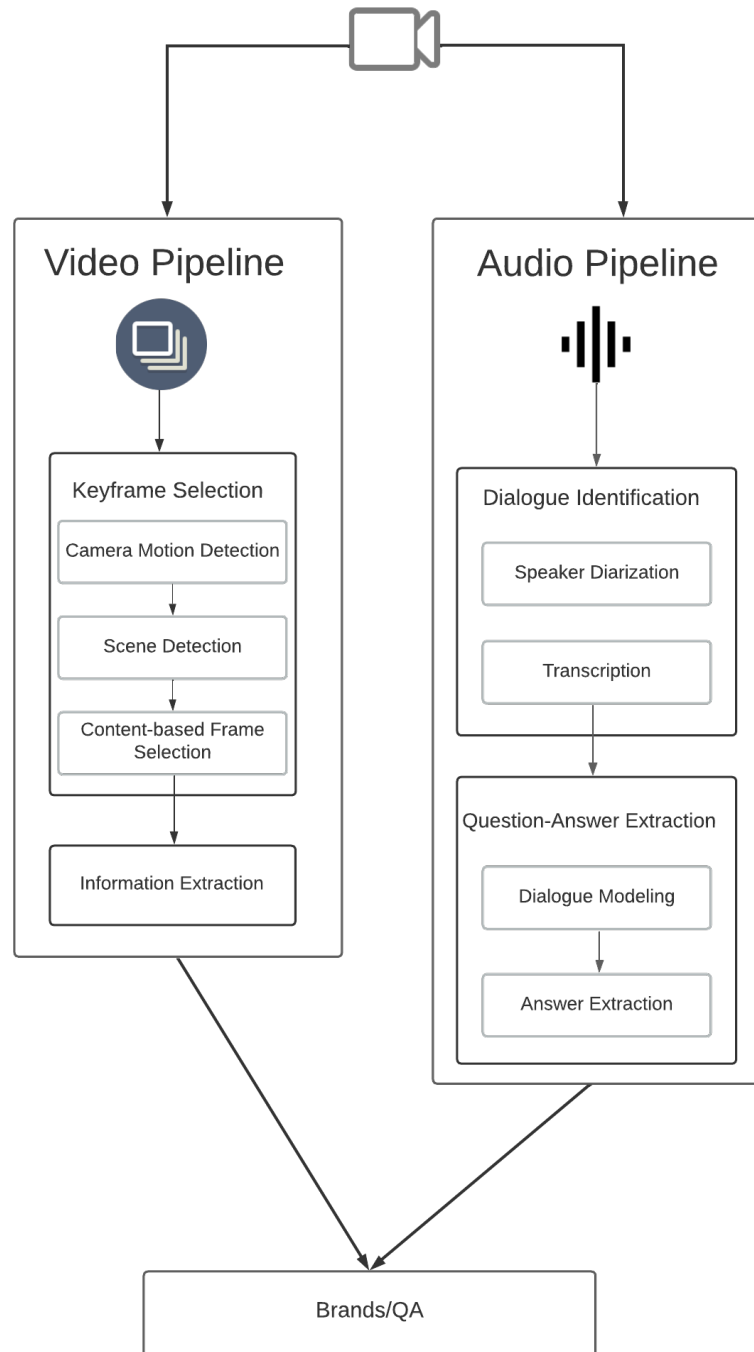


Figure 1.1: A diagrammatic of the two pipelines. The video pipeline analyzes video frames to extract inaccessible brands using a two-step process: key-frame selection and information extraction. The audio pipeline analyses extracts the question-answer pairs using a two-step process: dialogue identification and question-answer extraction.

In addition to these two pipelines, we also establish a third pipeline that harnesses both video and audio information in order to extract brand data. This pipeline specifically targets segments of the video that are directly relevant to the task at hand. By integrating video and audio analysis, we strive to create a more comprehensive and accurate approach for extracting valuable insights from the video content, ultimately leading to a deeper understanding of the subject matter.

Chapter 2

Related Work

2.1 Visual Accessibility

A substantial amount of research has discussed tools and systems to assist people with visual impairment. For example, VizWiz (5) is a mobile phone application designed to assist blind people in accessing visual information. VizWiz allows users to ask visual questions in nearly real-time, which are then answered by multiple human workers recruited through a general approach called quikTurkit. Applications such as Camfind and Taptapsee (26) utilize both volunteers and image recognition technology to assist visually impaired individuals. Alternatively, organizations like Be My Eyes offer free services to connect visually impaired people with sighted volunteers. This work is based on analyzing video conversations from users of Be My Eyes.

Already, one paper (3) has analyzed the usage of Be My Eyes. The authors conducted a survey with 30 blind users of the Be My Eyes mobile phone application. The survey aimed to identify situations where the application was useful, as well as areas of improvement and users' concerns. The survey found that most participants used the app between 2 and 5 times in short sessions between 1 and 3 minutes, and the app was most useful for reading text, finding things, and shopping assistance. However, the app received negative ratings for way-finding and helping at ATMs.

Qualitative responses revealed that users appreciated the app for working with non-accessible displays, matching clothing and makeup, and verifying the expiration date of food. Some users found the app less reliable due to operator delay and connection issues, while others had privacy concerns about sharing personal information with strangers (6). Suggestions for improvement included adding assistance category settings, improving the communication skills of volunteers through training, and making the app work at night.

Our work differs from the research presented as we analyze video conversations between visually impaired individuals and sighted volunteers to identify the specific brands and types of challenges they face in their daily lives, while the mentioned research focuses on evaluating the effectiveness of existing tools and systems like VizWiz, Camfind, Taptapsee, and Be My Eyes for visually impaired individuals.

2.2 Key-frame Selection

A body of work has explored how to address the high cost of analyzing a vast number of frames in videos by only analyzing select frames.

In (52), the authors introduce a way to make the tedious task of manually segmenting and labeling objects in video sequences more efficient. They tackle an active frame selection problem, in which they choose a set of 'k' frames for manual labeling that will result in the least amount of error in the automatic pixel-level label propagation process. Their proposed solution is based on a flow-based random field propagation model and uses a joint frame selection criterion and minimal manual annotation and pixel correction costs. The authors also develop an efficient dynamic programming method to optimize this criterion. The results show that their approach saves hours of human effort per video and performs better than several other methods.

In (17), the authors present a technique for automatically extracting a key-frame from a video sequence for use in searching videos on the internet. The technique

involves three steps: detecting shot boundaries, selecting shots, and extracting key-frames within the selected shots. The selection is based on measures of motion and spatial activity and the likelihood of people being present, determined by skin color and face detection. Simulation results on a large set of internet videos, including movie trailers, sports, news, and animation, show the effectiveness of the method, achieved at a low complexity cost.

In (4), the authors propose an object-based approach to extract key-frames that summarize the important visual content of videos. The method uses spatial segmentation of each frame to detect significant events, and each key-frame must present an important event such as the appearance and disappearance of relevant objects. The proposed method aims to capture the semantic content of a video shot while minimizing redundancy and memory space requirements. The video is split into shots, within which they look for events that involve significant objects appearing or disappearing, and select frames from those events as the key-frames. To figure out which objects are significant, the authors use a technique called "fuzzy coarse region segmentation" on the first frame of each shot, which basically means they separate the image into different objects based on their color and shape. They then pick out the important objects based on where they are in the frame and what they look like and use a model that takes into consideration the color histogram to match them between frames. The many-to-many correspondence between objects allows for handling complex interactions and occlusions between objects, and the comparison of each frame with already selected key-frames avoids redundancy in the final key-frames. This allows them to handle complex interactions and objects getting in each other's way and makes sure they don't pick the same frame twice. This is carried out for each frame until the end of the shot and each frame is compared to other key-frames from earlier shots.

Our main objective is to efficiently extract inaccessible brands from videos using a specialized pipeline. To achieve this goal, we have incorporated ideas from relevant literature, including techniques to reduce time complexity and processing costs, as

discussed in the mentioned work, which focuses on analyzing select frames to address the high cost of analyzing a large number of frames in videos.

2.3 Video Classification

In (53) the authors present a video representation technique using dense trajectories and motion boundary descriptors. Dense trajectories ensure good coverage of motion information in the foreground and surrounding context. Features aligned with trajectories are extracted for shape, appearance, and motion characterization since trajectories are able to track a spatial point over time. The introduced descriptor based on motion boundary histograms (MBH) outperforms other descriptors, especially for real-world videos with camera motion. The video representation is evaluated for action classification on nine datasets, and it consistently outperforms current state-of-the-art results.

In (12), the authors present a system developed for the Google Cloud YouTube-8M Video Understanding Challenge, which involves multi-label classification on the YouTube-8M dataset. The system uses various techniques to aggregate frame-level feature representations and generate video-level predictions such as recurrent neural networks (RNN) and generalized VLAD, frame-level features extracted using CNNs, and fusion strategies to explore complementarity amongst the models.

In (49) the authors present a new method for detecting human activities in untrimmed videos using untrimmed classification. The proposed method predicts the top 'k' labels for each untrimmed video based on global video-level features and generates temporally trimmed activity proposals using frame-level binary classification combined with dynamic programming. To obtain global video-level features, two methods are employed: one involves extracting features from the intermediate layer of ImageNetShuffle, while the other utilizes Motion Boundary Histogram as a feature extractor with a one-versus-rest SVM classifier. For frame-level features, a 3D-CNN is used to extract them.

In (45) the authors present an overview and analysis of the system developed for the untrimmed video classification task in ActivityNet Challenge 2016. The system utilizes multiple spatio-temporal clues such as frames, motion, and short video clips with CNNs and studies different quantization methods. It also uses improved dense trajectory with fisher vector encoding and MFCC audio features and OCR in the presence of audio or text respectively. The activities are classified using late fusion of one-versus-rest linear SVMs and OCR is employed for refining prediction scores.

Our approach does not rely on video classification techniques but instead draws inspiration from relevant works to incorporate some ideas and methods. Specifically, we utilize optical flow and dense trajectories to estimate the motion within a video as a subpart of our pipeline.

2.4 Visual Question Answering

In (25), the authors conduct a survey on visual question answering (VQA), which involves combining natural language processing and computer vision. They provide an overview of different datasets and models used in VQA and compare the datasets based on common factors. The models are classified into four categories, including non-deep learning models, deep learning models without attention, deep learning models with attention, and other models. The non-deep learning models involve Bayesian models, while the deep learning models without attention use CNNs to extract image features and RNNs to extract question features. The answer is decoded as either a classification task or a generation task. The attention-based models use architectures that assign weights to different parts of an image based on the question.

The authors of (48) provide a discussion on the fundamental concepts of VQA and address challenges with VQA methods, such as biased data and poor image quality. The authors highlight the need for application-specific datasets since current models trained on existing datasets may not be suitable for real-world applications, such as assisting visually impaired individuals. The authors summarize the different

approaches used for VQA, indicating that the first stage involves extracting image and question features, followed by training with attention-based mechanisms, including single-hop and multi-hop attention, as these models have shown better performance than others.

Complementing prior work, our pipelines extract information about inaccessible brands and gain insights into the issues that visually impaired individuals face by analyzing the conversations with visual questions asked by visually impaired users to sighted volunteers.

2.5 Dialogue Modeling

In (11) the authors address the problem of summarizing emails by proposing a graph-based approach to capture the conversation structure among emails and three cohesion measures (clue words, semantic similarity, and cosine similarity) to weigh the edges in the graph. The authors develop two different graphs - a fragment quotation graph, which essentially captures the conversational relationship at the sentence level, and a sentence quotation graph that helps distinguish between sentences that are more relevant to a conversation. The authors also propose two graph-based summarization methods, Generalized ClueWordSummarizer and PageRank, which are then applied to extract sentences as summaries.

In (31), the authors focus on recognizing dialogue acts (DA), which are labels assigned to individual utterances in a dialogue to indicate their function, such as questions, statements, or hesitations. Recognizing DAs is considered an important step in understanding dialogues. The authors compare two approaches: one that models the structure of utterances and another based on n-grams. They find that the sentence structure-based approach performs better than the n-gram-based method and that combining it with prosodic models further improves DA recognition accuracy. The study concludes that sentence structure-based approaches are more effective, especially when the DA corpus is small.

In (29) the authors explore different unsupervised approaches for modeling dialog acts. The authors investigate three models: a graph-theoretic framework, an HMM model, and an HMM+Mix model. The authors train and test these models on the temporal and graph-structural order of the datasets. The graph-theoretic framework involves forming a complete similarity graph, where the nodes represent sentences in a conversation and the edge weights represent the similarity between them. The goal is to cluster similar sentences together and separate dissimilar ones using a graph-partitioning approach called N-mincut. The HMM model assumes that a conversation is a sequence of hidden dialogue acts that produce observed sentences. The features of sentences, such as bag-of-words, author/speaker, position, and length, are modeled as multinomial distributions since they have discrete values. The HMM + Mix model essentially helps avoid unwanted topical clusters which are extracted by the previously said models.

In (51) the authors propose a statistical approach to modeling dialogue acts in conversational speech. They utilize lexical, colloquial, prosodic cues, and discourse coherence to analyze the structure of a conversation as a hidden Markov model. To model constraints on the sequence of dialogue acts, the authors use dialogue act n-grams. Additionally, they combine statistical dialogue grammar with word n-grams, decision trees, and neural networks to model the lexical and prosodic characteristics of each dialogue act. The authors also develop a probabilistic integration of speech recognition with dialogue modeling to enhance both speech recognition and dialogue act classification accuracy.

Our objective with the audio pipeline is to extract question-answer pairs by analyzing the conversation between the visually impaired user and the sighted volunteer. To achieve this, we utilize dialogue modeling in our pipeline to transform unstructured conversations into structured dialogues as an initial step to extract the question-answer pairs.

2.6 Audio-Video Fusion

In (55), the authors came up with a way to improve visual question answering for Audio-Video Scene-Aware Dialog (AVSD), which is a more complicated task than regular visual question answering because it involves multiple modalities. They created a method called QUALIFIER that uses a self-attentive multi-modal fusion module to combine different features and prioritize the question feature for generating a natural language response. To extract the video features, they used an I3D network that utilizes optical flow features, and for the audio features, they used a VGGish network since audio is a 2-dimensional temporal signal. Dialogue history features were extracted through hierarchical LSTMs, where the input to the LSTM is a question-answer pair. They tested QUALIFIER on a big AVSD dataset called DSTC7 and discovered that it performed better than other baseline methods. In conclusion, their proposed network architecture is effective in enhancing AVSD practice.

In (33), the authors propose a universal multi-modal transformer that addresses two challenges of audio visual scene-aware dialogue: 1) effective interaction between different modalities, 2) effective understanding of dialogues and generating informative responses. In order to extract video features, the authors use I3D-rgb and I3D-flow to extract RGB features and optical flow features. They use the VGGish model to extract audio features and for text features, they follow GPT2 and tokenize the input sentences into WordPieces. The authors use a self-supervised learning method and adopt multi-task learning (response language modeling, video-audio sequence modeling, and caption language modeling) approach to learn joint representations and generate informative and fluent responses.

In (19) the authors present a new approach to the audio-visual event localization task, which involves fusing information from multiple modalities effectively. They propose a joint attention mechanism that combines multi-modal fusion methods to improve the process. Their architecture learns representations from multiple modalities jointly, combining visual and auditory features and turning them into

joint representations. These representations are used to attend to visual and auditory features separately, producing new features that benefit from each other. The audio features are extracted using bi-directional LSTMs since this maintains temporal cues among the sequential stream. The video features are also extracted using LSTMs and to prevent loss of information, the authors use a scaled dot-product of audio and visual features for each segment to reduce the dimension of the raw visual features. The joint co-attention unit can be performed multiple times to obtain better joint representations.

The stated work differs from our objective in terms of scope. Through our combined pipeline we focus on brand extraction in videos by combining the audio and video pipelines to minimize the randomness of frame selection. The stated work focuses on various network architectures in the area of audio-visual event localization and addresses challenges specific to those tasks.

Chapter 3

Dataset Analysis

We now describe the analysis of our dataset, with the objective of identifying patterns concerning accessibility challenges and possible solutions. The dataset consists of a variety of videos that offer in-depth information on products or scenarios that are inaccessible to people with visual impairments.

3.1 Visual and Audio Content

We use thematic analysis to examine a dataset of videos that cover situations that are inaccessible to visually impaired individuals. Thematic analysis entails identifying, analyzing, and reporting patterns (themes) within data (8). One of the benefits of thematic analysis is that it allows researchers to organize and describe their data in rich detail, providing a comprehensive overview of the key themes that emerge from the data. This type of analysis can also help to identify gaps or inconsistencies in the data, highlighting areas where further research may be needed. By conducting this type of analysis, we aim to uncover themes such as the types of products or situations that are most commonly inaccessible, the challenges that visually impaired individuals face in accessing these products or situations, and potential solutions for improving accessibility.

In order to gain insight into the challenges faced by users in our dataset, we utilized an existing coding scheme (15) and modified it to fit our specific needs. Our dataset consists of 97 videos, each containing one or more questions related to visual representations of objects. The coding scheme we used involved four distinct indicators, each of which aimed to capture specific aspects of the videos and the questions they contained. These indicators consider both audio and video, and in some instances, the combination of the two. The first indicator, Indicator 1, focused on whether a video was legible or not. If the video was deemed legible, we proceeded to code the video using the remaining indicators. Indicator 2 was then used to assess whether we were able to decipher the task at hand based on the video alone, without needing any additional information from the audio. Indicator 3 focused on whether the object in question was clearly visible and in the foreground of the video, which would make it easier to answer the user’s question. Finally, Indicator 4 assessed whether the relevant areas of the object in question were present in the video. We use the same protocol as Davis et al, 2020, in our coding scheme. For the first part of Indicator 4, we evaluated each video to determine whether the relevant areas required to answer the question were present in the video alone. If so, we assigned the video a score of 1. If not, we assigned a score of 0. In cases where we were unsure, we assigned the video a score of Q, indicating that further analysis was needed to determine whether the question was answerable based on the video alone. In such cases, we reassessed the video in conjunction with the question to determine whether it was answerable.

After conducting the initial round of coding for each video in our dataset, we were able to identify several distinct issues related to both the video and audio components. Specifically, we identified three issues related to the videos themselves and two issues related to the audio or question components of the videos.

3.1.1 Visual Related Issues

Challenge 1: Extra Information

The main challenge in this dataset is to determine the answer based on the video since the extra information present is not relevant to the question. For example, when a user asks about the calorie count of a product, the video may contain information about the calorie count as well as information about ingredients, cooking instructions, etc., making it difficult to answer the question solely by analyzing the video.

Challenge 2: Poor video quality

We also encountered a significant challenge when attempting to discern the contents of some videos due to poor video quality. This was often the result of limitations in the hardware or equipment used by the users, as well as issues related to bandwidth and internet connectivity.

When videos are of poor quality, it can be difficult to accurately identify the objects or other relevant details presented in the footage, which can in turn impede our ability to analyze the video accurately. Addressing these issues requires a multi-faceted approach that may involve improving hardware and equipment, addressing bandwidth and connectivity issues, and developing more effective strategies for processing and analyzing low-quality video data.

Challenge 3: Information obscurity

A third significant challenge we faced in analyzing the videos in this dataset was the difficulty in identifying relevant information, which was often hindered by quality issues. Some videos are occluded, blurred, or have inadequate lighting, making it difficult to comprehend the video’s contents. Additionally, the relevant areas were frequently obscured, zoomed in to an extreme degree, or partially out of frame, making it difficult to detect them.

In some cases, the relevant areas were not even present in the video, which made it impossible to analyze the video and extract information from it. These issues underscore the importance of carefully considering lighting and orientation when filming videos, as well as ensuring that all relevant details are captured in the footage.

3.1.2 Audio related issues

Challenge 4: Poor audio quality

A considerable amount of the videos in the dataset had poor audio quality, which made it exceedingly difficult to understand the questions or tasks being asked. Yet, comprehending the questions is critical, as the visual content alone is often insufficient to answer the queries accurately.

Challenge 5: Unfavorable questions

Several users posed unclear and complicated questions, which made it challenging for volunteers to provide straightforward answers. Some questions were irrelevant to the video content, such as inquiries about the current date or the volunteer’s experience. Additionally, certain questions were ambiguous, such as asking for a recipe from a page that contained multiple recipes.

3.1.3 Variety of audio

We conducted a further manual analysis of approximately 50 videos and the corresponding audio. Our findings indicate that most of the audios commence with a greeting or some form of small talk, which may also persist after the question is posed. The majority of the questions begin with "Can/could" and "What." Some videos also contain a brief introduction of the task at hand before the question is asked. However, a few videos do not include any questions and the volunteers are able to state the apparent facts, making them unanswerable.

In our analysis of the types of questions asked, we found that a significant number of users inquire about reading text such as labels, cooking instructions, ingredients, and recipes. Typically, these questions are phrased as "Could/can you read the label/instructions?". The second most common type of question is users asking about the identity of an object, including its color and specific details, which are often framed as "What is this?". It's worth noting that users rarely ask just one question about the task at hand, and the questions are usually part of a conversation between the volunteer and the user, as users often have follow-up questions. There are a few cases where the questions are more extended than usual, such as when searching for an item or navigating.

3.1.4 Variety of answers

During this step, we examine the range of answers to users' questions. We classify videos and audio that are incomprehensible as unanswerable. For answerable questions, the length of most answers varies based on the questions asked. For instance, if a user requests assistance with reading a recipe, the answer would be as long as the recipe. Additionally, answers come in different types, depending on the question asked. These may include "yes/no" responses, as well as comprehensive answers that are generally very detailed.

3.2 Accessibility Challenges and Brand Presence

To better understand the types of tasks visually impaired individuals were seeking assistance with, we conducted a preliminary analysis and categorized videos into six distinct task categories. The first category is "reading text." In this task, users request help reading a label on a product, a recipe, or cooking instructions. The second category is "navigation." In this task, users ask for directions within a building or on a street. The third category is "help with an appliance." In this task, users

ask for assistance with operating a washing machine or oven. The fourth category is "comparison/sorting/search/counting." This category includes various scenarios where users request help with tasks such as counting currency or comparing the color of two towels. Additionally, users may ask for assistance with searching for a particular item or sorting a shelf. The fifth category encompasses a combination of these tasks and is labeled "miscellaneous."

Results from this analysis are shown in Figure 3.1. In this analysis, we selected videos to undergo both the video pipeline and combined pipeline based on the following criteria. We chose videos that belong to categories that have some form of brand presence, specifically "reading text," "help with an appliance," and "object description." Within each category, we evaluated if the brand or logo was visible in the video. If it was not, we excluded the video. However, if the brand was visible, we then checked if the brand was mentioned by either the user or the volunteer.

We found that in the "reading text" category, 15 videos met the conditions and were selected to go through the video and combined pipelines. No videos from the "help with appliance" or "object description" categories met the criteria. In total, 15 videos were selected for the video and combined pipelines due to the presence and mention of brands.

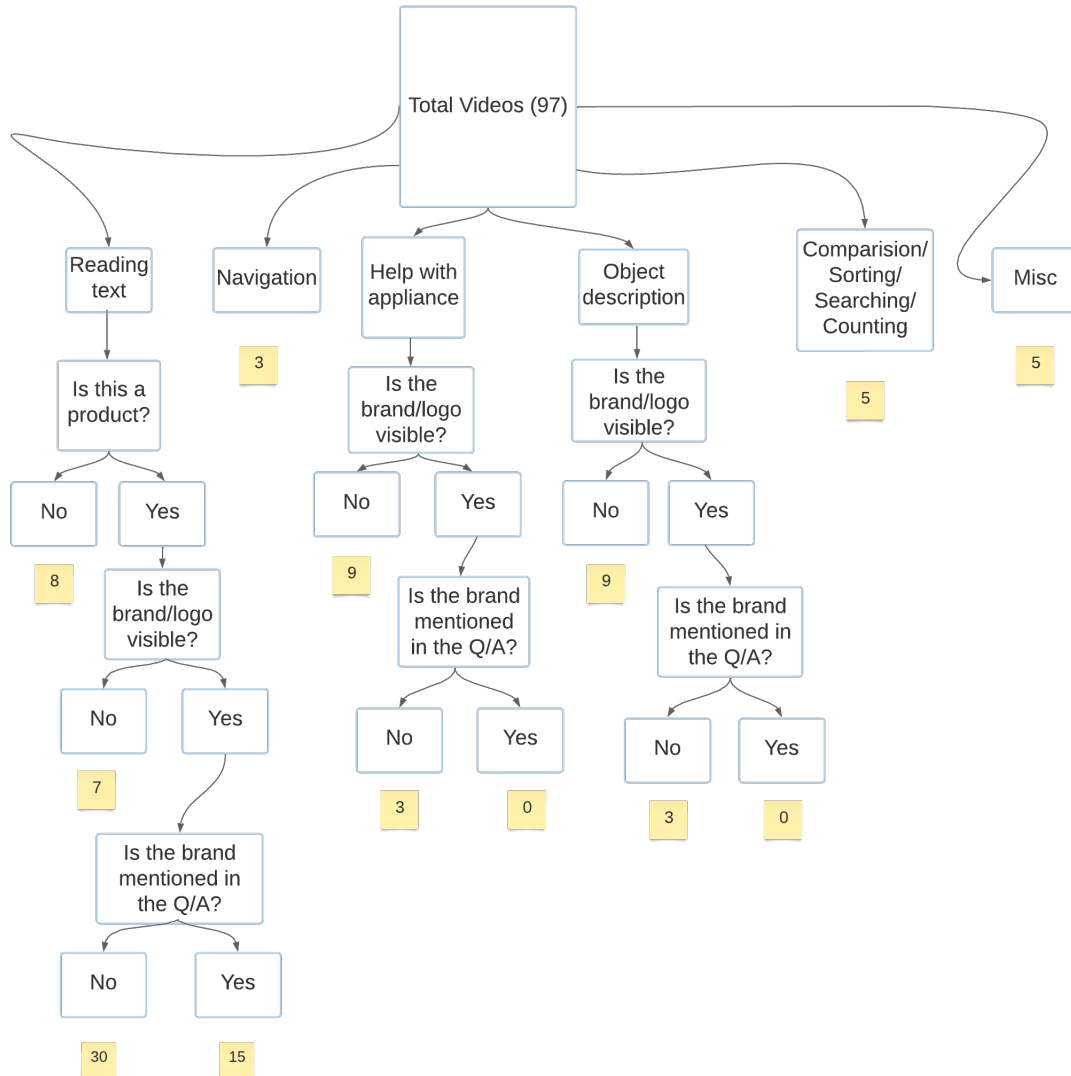


Figure 3.1: Flowchart depicting the categorization of videos. This analysis also aids in selecting videos suitable for analysis by the video pipeline for visual analysis.

Chapter 4

Video Pipeline

The video pipeline analyzes visual content to identify which brands are present, and so likely inaccessible. A key task is the identification of frames containing relevant content (18), followed by the analysis of this content to determine if there is any relevant brand information present. From our thorough review of the available videos, we found that 15 videos meet the precise requirements set forth by the pipeline of showing brand content (discussed in Section 3.2). We focus on these videos for this portion of the work.

The video pipeline consists of two main phases: key-frame selection and information extraction as shown in Figure 4.1. During the key-frame selection phase, frames are chosen that contain relevant information while frames that are redundant or unimportant are discarded. This process helps to make the analysis more efficient. Once the key-frames have been selected, important information is extracted from the key-frames. This information includes text, logos, or brands that may be present in the video, which can help identify brands (23; 42).

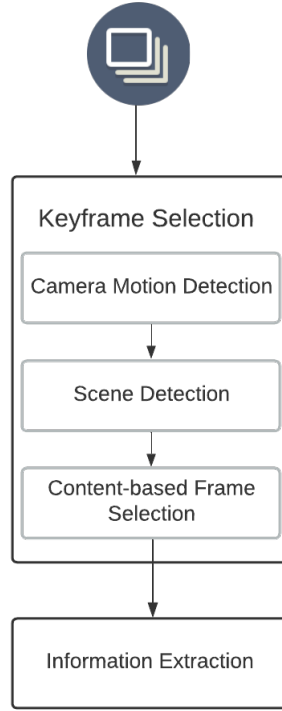


Figure 4.1: The video pipeline consists of 2 main steps: Key-frame selection and information extraction. The key-frame selection step is further divided into three sub-steps: camera-motion detection, scene detection, and content-based frame selection

Key-Frame Selection

We now describe our process for selecting key-frames from videos with important visual details. The video dataset that we are currently working with presents three unique challenges: First, there is a significant amount of camera movement present, generally panning motion. This movement is intentional and is caused by visually impaired users positioning their camera in a way that enables their volunteer helper to gain a better understanding of their surroundings. In addition, there are instances where the camera may remain fixed on a particular object or scene for an extended period, resulting in a series of frames that are similar in content. Finally, some frames may not contain any useful information, such as the absence of text, logos, or brand information. To address each of these challenges, we employ a three-phase approach

to identify key-frames: camera motion detection, scene detection, and content-based frame selection.

During the experimentation phase, we discovered that when the camera is in the process of panning, the amount of useful information that can be extracted from the frames is quite limited. We could use this to our advantage by estimating the motion of the camera and eliminating the frames associated with the panning motion. To accomplish this, we employed a technique known as camera motion detection, which involves extracting features from the dense optical flow between each pair of frames (22; 20). By analyzing the changes in the optical flow between each pair of frames, we are able to estimate the motion of the camera and determine which frames are panning or blurry. Once we have identified such frames, we eliminate them since they are unlikely to contain valuable information.

After removing panning and blurry frames from the video, the next step is to identify the scene boundaries, since this is a significant step in key-frame selection (39). This involves segmenting the video into a number of scenes, which can be achieved by detecting discontinuities in motion activity and changes in pixel value histogram distribution (17; 56; 28) using the pyscenedetect (Media) library in Python. After segmenting the video into numerous segments, we encounter a significant number of highly similar frames within each segment. This redundancy can be eliminated by randomly sampling 20% of each scene and analyzing this subset. This sample size was found effective in preliminary experiments. The quantity of frames sampled from each scene is determined by factors such as the size of the scene, the presence of brand or text information, and the amount of camera movement.

In an attempt to improve our sampling technique, we explored a non-randomized approach that involved using clustering algorithms to select frames from each scene (1; 32). The rationale behind this approach was to select a more diverse set of frames for analysis rather than relying on random sampling. To accomplish this, we used various clustering algorithms like community detection, K-Means, Hierarchical, and DBScan (34; 36) that were fed with image embeddings as inputs and then clustered the frames

based on their similarity. The aim was to select a fixed number of frames, n , from each cluster to analyze. The hope was that this would provide a more representative sample of frames from each scene, however, the results from this experiment were inconsistent across all the videos. One of the primary reasons for this was the variation in the clustering parameters that were used for different videos. As each video had a unique set of frames and content, the clustering algorithm’s parameters had to be fine-tuned for each video, which made the results unstable. Additionally, we have a specific need to choose frames based on their visual content, which further emphasizes the importance of selecting frames based on their content rather than a fixed set of parameters.

There are certain scenes where a considerable number of frames appear to lack any information. Analyzing such scenes is not only a waste of computational resources but also consumes a significant amount of time. To overcome this challenge, we apply a filtering mechanism that could quickly identify scenes that lack pertinent information. We specifically look for text or brand information (27) and, to accomplish this we use ”tester frames”, that are randomly sampled and constitute only 5% of the video frames. We put them through Microsoft’s text and brand detection APIs (14; 13). By obtaining a consensus from these tester frames, we can quickly determine whether or not the scene contains useful information. If the consensus is there is brand information, we proceed with further analysis.

After obtaining the consensus from the tester frames that a video scene contains valuable information, we proceed to analyze the video scene further by selecting 20% of its frames based on content. To achieve this, we utilize Microsoft’s text and brand detection APIs to identify frames that contain legible text, brands, or logos. By doing so, we ensure that the selected frames are visually informative and provide valuable context to the video’s content. Typically, the number of keyframes selected at the end of the process is significantly fewer, averaging 75 times less than the total number of frames.

Information Extraction

In our process of extracting information from the key-frames, we may sometimes encounter text and brand information that may not be directly related to the task of interest for the visually impaired individual. This can make the extracted information somewhat random in nature.

We address this issue by basing our information extraction model on the assumption that the camera is primarily focused on the object of interest. We aggregate the extracted information, which consists of the brand and text data that was obtained from selected frames using API calls, by using a combination of part-of-speech tagging and frequency count. Part-of-speech tagging helps us identify the various parts of speech present within the extracted text and brand data, we extract phrases that are nouns since we are specifically interested in identifying brands. Additionally, we use frequency count to identify the most commonly occurring words and phrases within the extracted brand and text data. This approach is effective because it allows us to filter out irrelevant information and focus on the most important information related to the object in question.

To improve the accuracy of our filtering of words and phrases, we attempted to utilize named entity recognition (NER) model (41). However, our initial results were less than satisfactory. The NER models we used were unable to correctly identify the extracted brands as entities, which led to the exclusion of potentially valuable information (47; 2). We believe that this issue stems from a lack of training data for the NER models pertaining to our dataset. The models we used were not specifically trained on the brands and organizations that are relevant to our dataset, which limited its ability to accurately recognize and extract this information. To address this, we would need to train a new NER model on a dataset that is more closely aligned with our particular domain.

Chapter 5

Audio Pipeline

The audio pipeline analyzes audio data obtained from the conversations between the volunteer and the visually impaired individual to identify the accessibility issues for visually impaired individuals. The goal is to retrieve the questions asked by the visually impaired individual and the corresponding answers from the volunteer.

The audio recordings in our dataset contain a sequence of dialogues between a visually impaired user and a sighted volunteer. Understanding the dialogues and extracting sets of questions and corresponding answers from these recordings requires overcoming several challenges that constitute a dialogue modeling problem. One challenge is the need to convert the transcribed audio into a sequence of dialogues, and this is done by identifying which speaker spoke at a given time and what was said. This is a crucial pre-processing step that serves as the foundation for our question-answer extraction model. In addition to speaker identification, another challenge we face is identifying the questions asked and modeling the conversation based on the speaker and questions. To address this, our question-answer extraction model assumes that either speaker may ask a question and the other speaker will provide the corresponding answer. Essentially, we divide the audio into various segments based on the speaker and dialogue type in order to accurately identify and extract the questions and answers from the conversation.

The audio pipeline is composed of two primary phases as shown in Figure 5.1: dialogue identification and question-answer extraction. The first phase, dialogue identification, employs a sophisticated diarization model that identifies and distinguishes between speakers (44) and converts the unstructured conversation into dialogues. The diarization model works by analyzing various acoustic and contextual features of the speech of each speaker. This process enables the model to identify who is speaking and what they are saying accurately. After converting the conversation into dialogues, the next phase of the model involves feeding them into the question-answer extraction process.

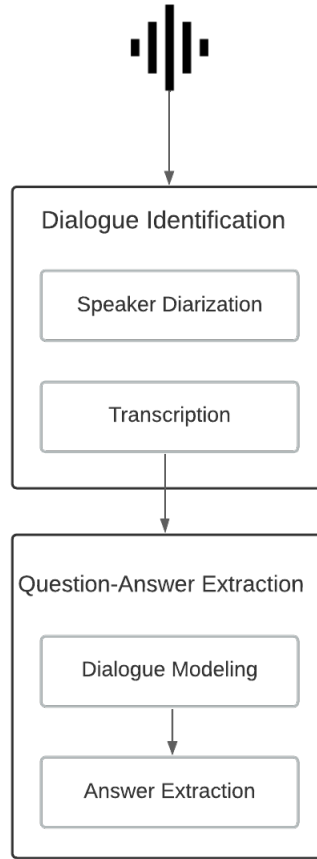


Figure 5.1: Diagram showcasing the audio pipeline’s two main steps: dialogue identification and question-answer extraction. The dialogue identification step includes speaker diarization and transcription. Meanwhile, the question-answer extraction step is comprised of dialogue modeling and answer extraction.

Dialogue Identification

To extract information from the unstructured conversation, we convert it into a structured dialogue using speaker diarization. We use the `pyannote.audio` library (10; 9), which provides an easy-to-use interface for working with audio data. This library allows us to apply a diarization model to the audio, which splits the audio into segments based on the speaker. Each segment corresponds to a particular speaker and indicates the duration of time that the speaker spoke.

Once we have identified the segments corresponding to each speaker, we transcribe them using OpenAI’s Whisper transcription model. We experimented with Microsoft’s transcription API as well, but found that the results were subpar, especially for short audio segments. The Whisper transcription model, on the other hand, appeared accurate and reliable also for short segments of speech.

Using the transcription output, we extract the conversation in the form of text where each dialogue is tagged to a specific speaker. This enables us to create a more structured representation of the conversation, which makes it easier to identify the questions and corresponding answers. We use this structured representation as input to the question-answer extraction module.

Question-Answer Extraction

The ultimate goal of the question-answer extraction process is to identify and extract sets of question-answer pairs from audio conversations by analyzing the corresponding transcribed text. This involves breaking down the process into two distinct steps: dialogue modeling and answer extraction. By carefully executing each step, we can obtain a better understanding of the conversational flow and identify the specific questions and answers that are being exchanged between the speakers. The resulting question-answer pairs can be used to identify patterns and trends in the conversation, which gives us a better understanding of the needs and concerns of the visually impaired user.

Dialogue modeling involves creating computational models that can understand and generate human-like conversations. The goal is to extract question-answer pairs from a conversation. Over the years, significant progress has been made in dialogue modeling, which can be achieved either in a supervised or unsupervised fashion. The task of dialogue modeling involves classifying dialogues into different categories such as questions, statements, apologies, and agreements, and then extracting patterns based on the dialogue labels. However, the supervised dialogue modeling systems (16) require training data and manual effort, and utilize predictors such as SVM-hmm (37), classifiers like Adaboost (30), Naive Bayes (24), hierarchical recurrent neural networks (46), and various other deep learning techniques (50).

On the other hand, unsupervised methods such as CRF (43), Markov Random Fields (21), Hidden Markov Models (7; 29), and graph-based (11) approaches have been used to address this issue. However, due to the scarcity of data, developing and training a model for dialogue modeling is not practical. Therefore, we can borrow from the above methods, specifically graph-based approaches, to develop a state machine that models dialogues.

Graph-based approaches are useful as they can model the different dialogue acts as nodes in a graph and the transitions between the nodes as edges in the graph. This allows for the development of a state machine that can model the entire conversation where every dialogue is classified as either a question or a statement. Such an approach does not require any training data and can be used to model different types of dialogues, and is especially helpful for our use-case, since we have two distinct types of dialogues. By using graph-based approaches for dialogue modeling, we can identify the underlying structure of the conversation and extract patterns that can help us understand the intent and meaning of the conversation.

The first step of dialogue modeling is to classify the dialogue as either a question or a statement. To do so, we employ a question classifier (35; 40) that classifies the diarized transcribed audio as either questions or statements. This step helps label each dialogue, which serves as input to the state machine. It is important to note

that in most cases, no questions are asked by the volunteer, but in some cases, they may ask questions to confirm whether the user got everything they needed or may ask questions that are related to positioning the camera to achieve a clearer view. Through the use of regex and phrase matching in our present classification method, we ensure that queries pertaining to camera movement are not labeled as questions, as it is likely that these types of inquiries are posed by volunteers. This process is essential for further analysis of the conversation since it divides the conversation into smaller segments, which enables us to determine both the timing of when a question was asked and the content of the question itself.

After classifying the dialogues as questions or statements, the next step is to model the dialogues using a state machine. This model helps extract the question and corresponding context spoken by the other speaker. For instance, if ‘Speaker A’ asks a question and ‘Speaker B’ responds to that question, this constitutes a question-context pair.

When it comes to dialogue modeling, it is important to have a clear understanding of how the different states in a conversation relate to one another. In our case, we have designed a state diagram that consists of six distinct states, each of which is characterized by a specific combination of the speaker and the type of utterance, which can either be a question or a statement.

In order to capture this variation in the conversation, each state in our state machine has four distinct edges, which correspond to the different possible combinations of the speaker and the type of utterance. Specifically, we have edges for speaker 0/question, speaker 1/question, speaker 0/statement, and speaker 1/statement, which allow for a wide range of possible transitions between states depending on the speaker and the type of utterance.

This state machine as shown in Figure 5.2 allows us to accurately extract relevant question-answer pairs from the conversation, as it takes into account the different ways in which the conversation can unfold. By modeling the conversation as a state

machine, we are able to capture the dynamic nature of the dialogue and ensure that our model can handle a wide range of inputs and scenarios.

To gain a deeper understanding of the state machine, consider a basic conversation between two speakers where one speaker asks a question and the other speaker provides a response.

At the beginning of the conversation, the current state is determined by whether a question is asked and who asks it. If a question is asked, the current state could be state 0 or state 2, depending on the speaker. If no question is asked, the current state could be state 4 or state 5, also depending on the speaker. The goal of extracting question-context pairs requires first locating the question. Once the question has been extracted, the next state becomes either state 1 or state 3, which helps extract the context, depending on who asked the question. For example, if speaker 0 asks a question and speaker 1 answers, the next state would change to state 1 and remain in that state until another question is asked by either speaker. If no questions are posed during the conversation, the machine alternates between state 4 and 5 until the conversation is completed.

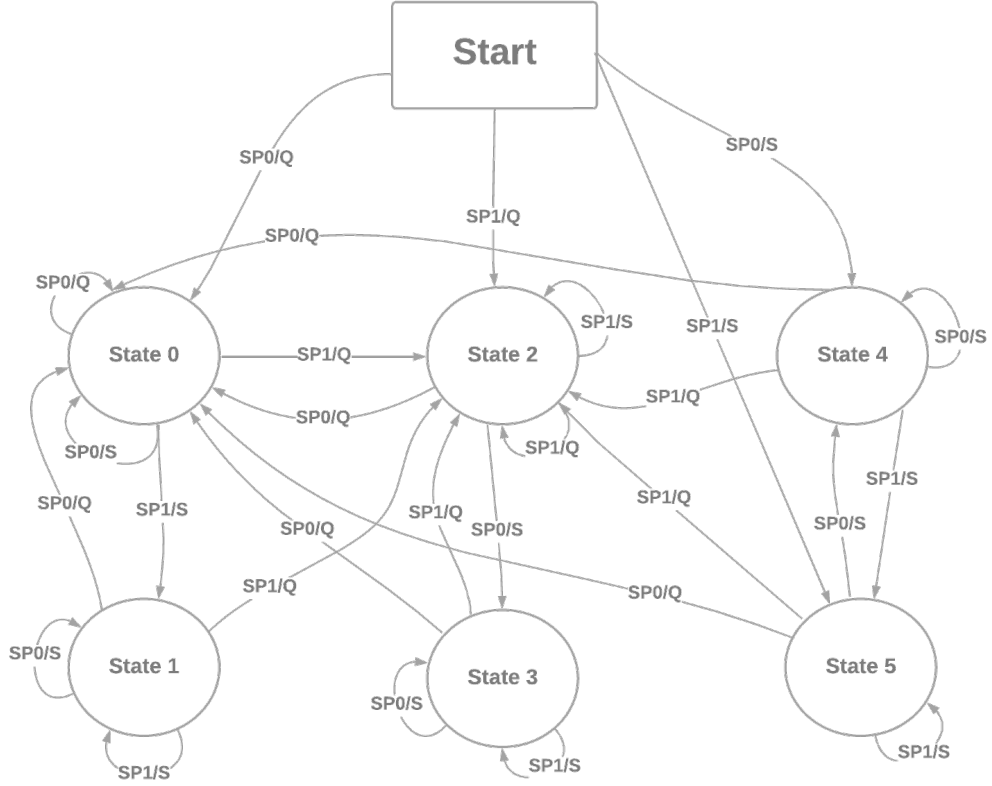


Figure 5.2: State Machine depicting the conversational dynamics between two individuals, illustrating the question-and-response pattern between the speakers.

Once we have modeled the dialogue and identified the question-context pairs in the audio conversation, the next step in the question-answer extraction process is to extract the answers from within the corresponding context. This step requires natural language processing techniques to be applied to the context in order to identify the relevant answer. To achieve this, we make use of HuggingFace’s Roberta QA Model, which is a pre-trained deep-learning model specifically designed for question-answering tasks. This model takes the question and context pair as input and uses contextual embeddings to mine the answer from within the context with respect to the given question. The Roberta QA Model is particularly effective because it has been trained on a large amount of textual data and is capable of identifying subtle contextual cues that indicate the correct answer.

Chapter 6

Combined Pipeline

As our dataset contains both video and audio modalities, we are interested in developing an approach that can effectively leverage both modalities to improve our analysis of the videos. To achieve this, we draw inspiration from the research discussed earlier, which explores various deep-learning techniques for multi-modal fusion.

As previously mentioned, the video pipeline involves randomly selecting key-frames throughout a video, which allows for the identification of certain brands. However, this method can also result in the inclusion of unrelated information due to the presence of random objects or text in the background. Furthermore, there may be instances where the brand cannot be identified if the frames do not provide a wrong consensus.

Given the stochastic nature of the video pipeline, efforts are made to minimize the randomness of frame selection by combining the video and audio pipelines. The audio pipeline allows for the extraction of information regarding the questions asked, the answers provided, and the timing of each. By conducting a targeted search within the video using this extracted information, key-frames that are associated with the brand are selected.

In order to merge the video and audio pipelines, we implement a process that involves two primary stages as referred to in Figure 6.1: sub-clip selection and brand extraction and validation.

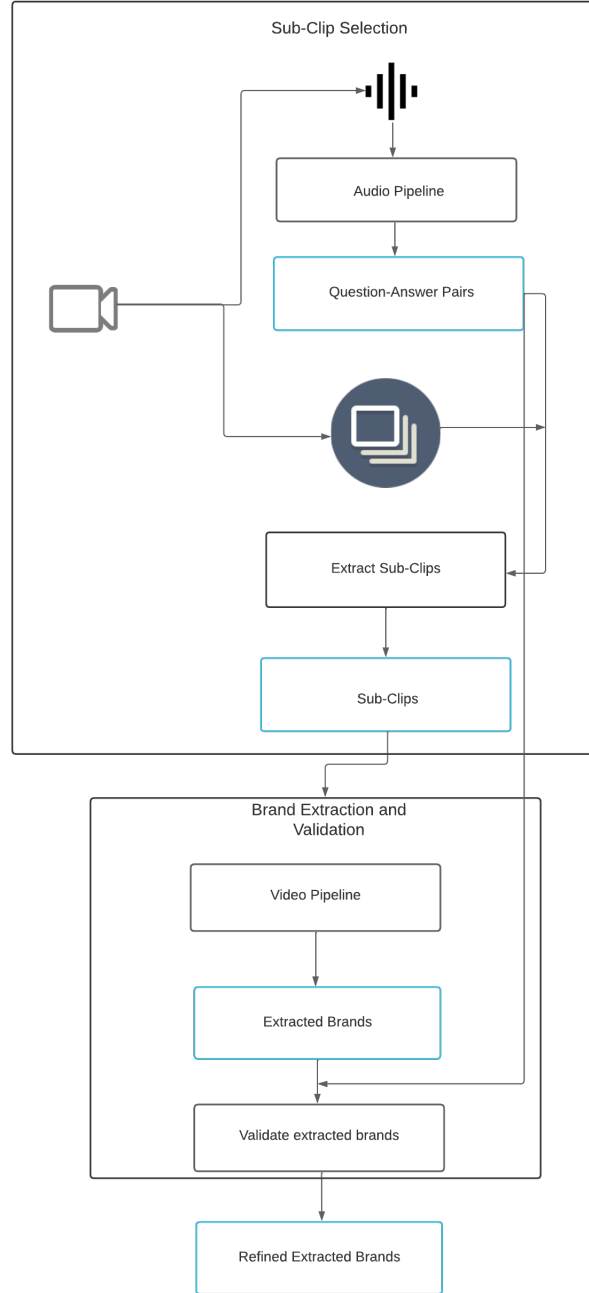


Figure 6.1: Diagram showcasing the combined pipeline’s two main steps: sub-clip selection and brand extraction and validation. The sub-clip selection step includes extracting sub-clips from videos based on extracted question-answer pairs. Meanwhile, brand extraction and validation are comprised of extracting brands from the sub-clips and validating them with the question-answer pairs.

The first stage of sub-clip selection involves choosing specific segments from the video that align with the question-context pairs that were extracted from the audio pipeline. To accomplish this, we follow a two-step approach. First, the audio is sent through the audio pipeline to extract the question-answer pairs. Then, using these pairs and the corresponding frames, we extract the sub-clips. This is because the question-answer pairs help us identify the specific moments when a question was asked and answered, allowing us to select the most relevant parts of the video footage. This is a critical step, as it enables us to obtain a relevant and accurate representation of the information that is being analyzed.

After the selection of sub-clips, we move on to the next stage which involves analyzing and extracting information from each sub-clip. To achieve this, the sub-clips are passed through the video pipeline. Within the video pipeline, the brands present in the sub-clips are extracted. However, to support the accuracy and reliability of the extracted information, a validation process is carried out. This validation process compares the extracted brands with the entities present in the question-answer pairs, ensuring that the brands extracted are indeed relevant to the context of the video. The goal of this analysis is to identify key entities and information within the video sub-clips that are relevant to the questions and contexts extracted from the audio pipeline.

The integration of both pipelines provides the added benefit of validating the outcome of the video pipeline with that of the audio pipeline, thus reducing the likelihood of false positives. By combining these two pipelines, we are able to provide a more comprehensive and accurate analysis of the video content, which is particularly important in the context of brand analysis.

Chapter 7

Results

7.1 Quantitative Analysis for Brand Detection

The main objective of the video and the combined pipeline is to extract valuable information about brands from video data. To evaluate the effectiveness of our approaches, we developed a set of ground truth data for all 15 videos that fall within the scope of the video pipeline. The ground truth data consists of a comprehensive set of brand labels that cover all instances of brand exposure in the videos. For evaluation, since the task is brand detection, it is natural to use metrics relevant to other detection problems such as object detection. In order to quantify the performance of the brand detection models, we define the ground truth as a set of brand names (GT) and the model predictions as another set of entities (Pred). For instance, a typical set of ground truth and predicted results can be:

$$GT_1 = \{'Parle'\}$$

$$Pred_1 = \{'chocolate', 'granola', 'Parle'\}$$

The ground truth set (GT set) contains the brand "Parle", since this is the brand in question. The predicted set contains "chocolate", "granola" and "Parle" since

these entities were captured by the pipeline. To evaluate the overlap between these two sets, we calculate an Intersection over Union (IoU) score for each example. This score measures the similarity between the predicted brand names and the ground truth brand names and is computed using equation 7.1. In more detail, the IoU score is a standard metric used in object detection tasks that computes the ratio of the intersection between two bounding boxes (in this case, the sets of brand names) to their union. By applying this metric to the sets of predicted and ground truth brand names, we can evaluate the degree of overlap between them and obtain a measure of how well the model is able to detect the brands of interest. A high IoU score indicates a good match between the predicted and ground truth brand sets, while a low score indicates poor performance.

$$IoU = \frac{|Pred \cap GT|}{|Pred \cup GT|} \quad (7.1)$$

When determining the accuracy of predictions, we categorize them as either True Positive (TP) or False Positive (FP) based on a pre-defined threshold value, denoted as α , where α is 0.3. Predictions with an IoU score greater than or equal to α are classified as TP, indicating that the predicted set matches with the ground truth set. Conversely, predictions with an IoU score less than α are categorized as FP, meaning that the predicted set does not align with the ground truth.

In the context of detection, we do not have the concept of True Negatives (TN) since the ground truth set is always present, and there is no scenario where they are absent. On the other hand, it is possible for the prediction set to be empty, which we refer to as a False Negative (FN) scenario. In this case, the brand is present in the ground truth, but the pipeline fails to detect it, resulting in a false negative prediction.

Once we have categorized the predictions into the three categories mentioned above (True Positive, False Positive, and False Negative), we can then calculate the

Precision, Recall, and F1 score for the pipelines. These metrics are calculated using equations 7.2, 7.3, and 7.4, respectively.

The performance of these pipelines is assessed by considering precision, recall, and F1 score. Precision reveals the accuracy of the positive predictions made by the model, while recall indicates the extent to which actual positive instances were captured by the predictions. These metrics are evaluated from two different viewpoints. In one, precision is given greater weight, with limited tolerance for error. In the other, the focus is on correctly predicting the expected output, regardless of the number of false positive predictions.

$$Precision(P) = \frac{TP}{TP + FP} \quad (7.2)$$

$$Recall(R) = \frac{TP}{TP + FN} \quad (7.3)$$

$$F1 = 2 \cdot \frac{P \cdot R}{P + R} \quad (7.4)$$

Table 7.1: This table represents the quantitative results of the video and combined pipelines. The combined pipeline outperforms the video pipeline with higher precision, F1 score and lesser time complexity.

pipeline/metric	Precision	Recall	F1	Time Taken (seconds)
Video	0.57	1	0.72	333.5
Combined	0.82	0.75	0.78	17.50

When evaluating the performance of machine learning pipelines, it is important to take into account their precision and recall values, as well as their overall F1 score. In this particular case, we have compared the performance of a video pipeline with that of a combined pipeline. The combined pipeline is expected to be more accurate than the video pipeline because it performs a targeted search, whereas the video pipeline does not.

In order to assess the efficiency of each pipeline, we also take into account the average time each one takes to complete as referred to in Table 7.1. It’s clear that the combined pipeline is about 20 times faster compared to the video pipeline.

As a consequence of not performing a targeted search, the video pipeline is expected to produce more false-positive results than the combined pipeline. In other words, the video pipeline is more likely to identify brand names in situations where there are none, which would lead to a higher number of incorrect positive predictions. Moreover, due to this targeted search the combined pipeline is way faster than the video pipeline since it doesn’t have to look through the entire video, but rather select segments of it based on the output from the audio pipeline.

When we examine the experimental results, we observe that the recall value of the video pipeline is higher than that of the combined pipeline. This means that the video pipeline is more successful at identifying true positive instances, or actual brand names in this case. However, the video pipeline also suffers from a significant loss of precision, as it is more likely to make false positive predictions. This significant skew of the video pipeline towards recall is problematic because it decreases the overall F1 score, which takes into account both precision and recall.

In conclusion, although the video pipeline has a higher recall value than the combined pipeline, it suffers from a loss of precision that negatively impacts its overall performance. The combined pipeline, which performs a targeted search, is expected to be more accurate and produce fewer false positives.

7.2 Qualitative Analysis for Audio Pipeline

In order to assess the efficacy of our audio pipeline, we conduct a qualitative analysis of how well it extracts question-context pairs from the audio recordings. These question-context pairs consist of a set of questions that may be asked by either the user or the volunteer and the context is the response given by the other speaker to the question asked. Of note, we do not assess the question-answer pairs generated by Roberta’s

QA model, which mines information based on the given question and context. This is primarily because, for most tasks, there isn't a single, definitive answer to a particular question.

For instance, a series of extracted question context pairs:

Question 1: Hi, how are you today?

Context 1: I'm good, thanks for asking.

Question 2: I was wondering if you could tell me what this is.

Context 2: Um, yeah okay. Let me see. Turn it over once again.

Question 3: Can you see it or should I move it back?

Context 3: This looks like a packet of chips.

Question 4: If you could, uhh, could you tell me the brand and the flavor?

Context 4: The brand seems to be Kurkure and they are spinach chips.

To evaluate the effectiveness of the audio pipeline, we employ three key criteria. Firstly, we check whether the questions extracted from the conversation are the same as those actually asked in the audio recording. Secondly, we assess whether the contexts that have been extracted from the conversation correspond to the response to the questions that were asked in the audio recording. Finally, we check whether all of the question-context pairs have been extracted, assigning greater priority to questions that are directly related to the task at hand.

We obtained results for all 97 videos, each of which involves a different task such as reading text and navigation. To assess the performance of the audio pipeline based on the specified criteria, we categorize the results into three groups due to the subtle variations between the extracted results and the expected results. We specifically utilize these categories when evaluating the videos based on criteria 1 and 2. The first category consists of extracted results that match the expected results exactly. The second category encompasses extracted results that differ from the expected

results to some extent. The third category encompasses extracted results that are significantly different from the expected results.

Criterion 1: Are extracted questions the same as the questions posed?

One criterion examined was whether the extracted questions were the same as the questions posed in the videos. Out of the 97 videos analyzed, 37 belonged to the first category since they had extracted questions that were identical to the original questions posed. The example listed below illustrates the results belonging to this category.

- **Example 1:**

Extracted Question: What do you need help with?

Actual Question: What do you need help with?

Extracted Question: What are the cooking instructions for this? What should I set my microwave temperature to?

Actual Question: What are the cooking instructions for this? What should I set my microwave temperature to?

However, 45 of the 97 belonged to the second category since there were differences between the extracted questions and the actual questions. These differences ranged from minor issues such as transcription errors or capturing phrases spoken by the other speaker in the context. This category also includes instances where some questions within a conversation's series were extracted correctly, while others were extracted incorrectly. This discrepancy was likely due to errors in dialogue extraction or incorrect classification of the questions. For instance, in a video containing a series of questions, the extracted question and the actual question posed could appear as follows for videos that belong to the second category:

- **Example 2:**

Extracted Question: Is this yellow? Yep. Yep.

Actual Question: Is this yellow?

Extracted Question: I was wondering if you could tell me what this is? Um yeah okay. Let me see.

Actual Question: I was wondering if you could tell me what this is?

- **Example 3:**

Extracted Question: This is a shirt that's yellow, yes?

Actual Question: N/A

Extracted Question: I was wondering if you could tell me what this is?

Actual Question: I was wondering if you could tell me what this is?

The examples stated above depict a subset of the range of differences between the question extracted and the actual question posed. In example 2, the extracted questions contain additional phrases as a result of the incorrect classification of dialogues from the diarization module during dialogue extraction. In example 3, only one of two questions in a conversation was accurately captured. This occurs when the question is not asked in a proper format, causing the question classifier to classify it as a statement.

In an extreme case, 15 videos of the 97 belonged to the third category, where none of the questions in the conversation were extracted or were vastly different from the actual questions posed. This likely occurred due to the false classification of a statement as a question, incorrect extraction of dialogues, or false classification of a question as a statement. The examples below depict a subset of extracted questions for this category.

- **Example 4:**

Extracted Question: N/A

Actual Question: I'm trying to understand what this is.

- **Example 5:**

Extracted Question: The expiration? I'm trying to understand what this is.

Actual Question: The expiration date is?

In example 4, the question wasn't extracted since it was falsely classified as a statement. In example 5, the context was falsely classified as the question, leading to wrong extracted results. Overall, this criterion revealed variations in the quality of question extraction, which may have implications for the accuracy of the data analysis.

Criterion 2: Are extracted contexts the same as the contexts presented?

This analysis also examined whether the extracted contexts were the same as the contexts presented in the videos, as part of Criterion 2. Of the 97 videos analyzed, 46 had extracted contexts that belonged to the first category since they were identical to the actual contexts presented. The example listed below illustrates the results belonging to this category.

- **Example 6:**

Extracted Context: I believe what you're looking for is on the right.

Actual Context: I believe what you're looking for is on the right.

However, 32 video instances belonged to the second category where there exists a range of differences between the extracted and actual contexts, which could be attributed to transcription issues, and incorrect dialogue extraction. Additionally, in some videos, only some of the contexts within a series were correctly extracted, while others were incorrect or partially extracted, typically due to audio issues or incomprehensible speech.

- **Example 7:**

Extracted Context: That's ranch. Okay. Thanks.

Actual Context: That’s ranch.

- **Example 8:**

Extracted Context: That’s definitely in color

Actual Context: That’s definitely grey in color

Extracted Context: N/A

Actual Context: Sunny there? Have a good day

The examples stated above depict a subset where the extracted contexts differ from the actual context in varying degrees. In example 7, The context contains some extra phrases due to incorrect dialogue extraction. The diarization model is sensitive to short segments of speech, making it difficult to differentiate between speakers in such cases. Example 8 represents a case where the context wasn’t extracted properly due to bad transcription and the false classification of the statement as a question.

There are 19 videos that belonged to the third category where none of the extracted contexts were similar to the actual contexts. These cases were primarily attributed to multiple issues such as incorrect dialogue extraction, which in some cases merges the question and context into one statement, particularly in cases where the context was a single-word answer. This occurrence is common when the pitch and tone of the speakers are almost indistinguishable, making it challenging to distinguish between them. Additionally, the misclassification of statements as questions and vice versa is responsible for the incorrect extraction of contexts.

- **Example 9:**

Extracted Context: Spicy noodle soup. Okay, it’s probably. Thank you

Actual Context: Okay, it’s probably on the right towards the inside. It has a picture of a red tree.

Extracted Context: N/A

Actual Context: Maybe center it or position the camera differently. Ah, there we go. It seems to be turned towards the window.

In the example mentioned above, none of the extracted contexts were consistent with the actual context in a series of contexts. This was due to a combination of incorrect dialogue extraction, incorrect transcription, and incorrect classification of the context as a question. These findings highlight the need to carefully evaluate the accuracy of the pipeline used for extracting contexts from the videos.

Criterion 3: Have all the question-context pairs been extracted?

Another criterion evaluated the efficacy of the pipeline in extracting all question-context pairs related to the task. Specifically, the analysis aimed to assess whether all questions related to the task were extracted, which is a subset of all question-context pairs.

Out of the 97 videos analyzed, the pipeline was able to successfully extract question-context pairs related to the task for 40 videos. Additionally, the pipeline extracted all question-context pairs, including task-related pairs, in approximately 35 videos. For about 39 videos, some of the question-context pairs within a conversation were extracted, while others were missed, and for 23 videos, no question-context pairs were extracted due to the incorrect extraction of questions and contexts.

Overall performance

We were able to obtain satisfactory results for the majority of the questions, contexts, and question-context pairs. However, there were a few instances where questions, contexts, and question-context pairs in a series of questions, contexts, and question-context pairs were partially correct or missing. It's worth noting that the success of the pipeline heavily relies on the speech diarization process used to extract the dialogue. The accuracy of the diarization model can be affected by various factors, such as the length and complexity of the speech segments, the number of speakers

involved, and the similarity of their pitch. In cases where one of the speakers speaks in short phrases or if both speakers have a similar pitch, the diarization model may not perform as well, which can result in lower accuracy in the extracted question-context pairs.

Additionally, we found that the performance of the pipeline can also be affected by the classification accuracy of the question classifier. There were instances where some questions asked by users were not classified as questions due to the phrasing used, which can lead to missing or incorrect question-context pairs being extracted. Therefore, it's important to ensure that the question classifier is trained to accurately identify a wide range of question phrasings and structures.

Overall, while the audio pipeline produced satisfactory results, there are still areas where improvements can be made, particularly in enhancing the speech diarization process and refining the question classifier.

Chapter 8

Conclusion

The main aim of our research is to analyze videos in order to gain insight into inaccessible brands and a series of questions and corresponding answer pairs in order to further understand what issues visually impaired individuals face. Our research addresses the challenges of analyzing videos, specifically the high cost of evaluating numerous frames and the unstructured nature of conversations between visually impaired users and sighted volunteers. To tackle these challenges, we developed an approach that incorporates two separate pipelines for analyzing visual and audio components. The video pipeline identifies inaccessible brands, while the audio pipeline focuses on understanding the types of questions and answers discussed in the conversations.

Our work involved determining the relevance of videos for analysis, reducing computational costs by detecting key-frames, and developing a pipeline that can identify speakers and extract relevant information from unstructured conversations. Additionally, we faced the challenge of integrating video and audio data to extract pertinent brand information, which required the successful fusion of information from various sources while maintaining and leveraging the distinct attributes of each modality.

In assessing the performance of our individual pipelines, we carried out a quantitative analysis of the results from both the video pipeline and the combined pipeline. The combined pipeline demonstrated superior performance, achieving an F1 score of 0.78, primarily attributed to its higher precision. This indicates that the combined pipeline successfully extracted brands with fewer false positives, although it occasionally missed brands that were genuinely present. The reason for this performance can be attributed to the targeted search conducted by the combined pipeline throughout the video. While the video pipeline detected numerous false positives, it attained a recall score of 1, signifying that it captured brands from all the videos, along with the presence of false positives in the extracted data. To evaluate the effectiveness of the audio pipeline, we conducted a qualitative analysis of the extracted question-context pairs. In most instances, the pipeline accurately identified the question and context pairs, closely resembling the actual questions and contexts mentioned in the video. However, in a few cases, the pipeline failed to accurately extract these pairs due to challenges related to dialogue extraction or question classification.

In conclusion, we have successfully created two primary pipelines that effectively analyze videos and extract valuable information concerning brands and the challenges faced by visually impaired individuals. These pipelines demonstrate the potential for leveraging multi-modal approaches to gain a deeper understanding of the subject matter and to address real-world problems. By employing these methods, we can not only enhance the accuracy and efficiency of video analysis but also contribute to the development of solutions that improve the lives of visually impaired individuals and help them overcome daily obstacles.

Chapter 9

Future Work

There is considerable scope for future work in improving the automated pipelines for analyzing conversations between visually impaired individuals and volunteers.

One primary area in the video pipeline is the extraction of unnecessary information along with the brand name. Cooking instructions, calorie content, and ingredient lists are among the examples of extraneous information that get extracted and can hinder analysis. Therefore, future work can focus on improving the video pipeline to reduce the frequency of false positives.

Similarly, the audio pipeline can be improved to extract only those questions that are relevant to the task at hand. In the present study, the audio pipeline often extracts question-context pairs that are not pertinent to the task, and this can impede the efficiency of the pipeline. One potential solution to this issue is to employ natural language processing (NLP) techniques to better understand the semantic meaning of each sentence and extract only those sentences that are relevant to the task.

Moreover, to fully leverage the benefits of both modalities, future work can explore the development of more efficient ways to combine video features and audio features. One possibility is to ground the audio with respect to the video, which could help improve the accuracy of the extracted information. By grounding the audio with

respect to the video, it may be possible to reduce the frequency of false positives and enhance the overall accuracy of the analysis.

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